RAHUL MUTHU

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Education

Stony Brook University — Stony Brook, NY
B.S. in Computer Science & Applied Mathematics — Expected May 2027
Relevant Coursework: Data Structures, **Analysis of Algorithms**, **Software Engineering**, **Systems I**, **Systems II**, **Computational Geometry**, **Probability & Statistics**, **Deterministic & Stochastic Models**
Dean's List — Spring 2025

Experience

- **Teaching Assistant** — Jan 2026 – Present
  Stony Brook University — CSE320 — Systems II — Stony Brook, NY
  - **Authored** homework assignments, **autograder tests**, and **exam** questions for systems programming topics.
  - **Held** weekly office hours and review sessions, explaining debugging, memory, concurrency, and assignment design.
- **MicroBook** — Feb 2026 – Jul 2026
  Independent Systems Project — C++20 — Remote
  - **Built** a deterministic single-symbol **limit order book** with validated domain types, order-id lookup, snapshots, and invariant checks.
  - **Implemented price-time matching** for marketable limit orders with partial fills, full fills, maker-price trades, and deterministic trade order.
  - **Added** replay/verification CLIs with strict text parsing, final-state hashing, fixed-seed invariant tests, and **Google Benchmark** workloads.
- **Research Assistant** — Oct 2024 – Aug 2025
  Stony Brook University — Stony Brook, NY
  - **Developed** a modular **AutoAnnotator** in **PyTorch**, using grid search and k**-fold cross-validation** on **27k** cell profiles.
  - **Engineered** a **SLURM** training pipeline for **55 GB/day** across **8xA100 GPUs**, cutting epoch time **65%** (**130 min** → **45 min**).
  - **Implemented** hierarchical classifiers, class-balanced loss, and feature selection, improving rare-cell **F1** by **15 pp** and reducing features **23k** → **3.5k**.
- **Software Engineering Intern** — Jul 2024 – Aug 2024
  Giesecke+Devrient — Dubai, UAE
  - **Developed** an **XGBoost** demand-forecasting service using checkout data, reducing physical cash inventory requirements by **40%**.
  - **Deployed** a serverless ETL pipeline with **AWS Lambda** and **Kinesis**, cutting dashboard **p95** latency **50%** (**180ms** → **90ms**).
  - **Authored Jest** tests for a **TypeScript** analytics service with **90%+** coverage and automated releases using **Docker**, **Kubernetes**, and **GitLab CI**.

Projects

- **Hold'Em Engine (C++17)** — Poker UI for strategy modelling; **Monte Carlo** pot-equity simulations; **bitwise** hand evaluator
- **Plinko Simulator (p5.js)** — Interactive Plinko game analyzing ball dampening, X-biasing, odds, and player engagement

Activities

- **Founder & President – Stony Brook Quant Club**: 80+ members; **Jane Street Estimathon**, **Flow Traders** events
- **ICPC Qualifier**: 2nd overall junior category; 200+ problems (**Codeforces**, **LeetCode**, **AtCoder**)
- **UAE IMO National Qualifier**: Rank 6 nationally; **IMO** training camp invite
- **Stellantis Student Award**: €1,500 scholarship; **Stony Brook International Excellence**: $1,000/semester

Publications & Presentations

- B. Lukacsy, **R. Muthu**, A. Jalwan. *Supervised Deep Learning for Automatic Cell-Type Annotation with scRNA-seq*. Presented at the **URECA Undergraduate Research Symposium**, Stony Brook University, 2025

Technical Skills

- **Programming**: Python (**NumPy**, **pandas**, **scikit-learn**, **PyTorch**), **C++17/20**, **Java**, **SQL**, **Bash**
- **Systems**: **Linux**, **AWS**, **Docker**, **Kubernetes**, **SLURM**, **Git**, **CI/CD**